

Quantum-like mind: consciousness as an analogous system to quantum theory

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Abstract

Since the inception of quantum mechanics, the general consensus is that only fundamental particles - objects on the scale of $\hbar$ - operate within a quantum framework. However, as computing techniques advance, the study of complex systems has yielded patterns and results that make us question the restriction of quantum theory to subatomic scales. Quantum-like (QL) theory explores systems existing on the classical scale that may be modeled using the mathematics of quantum theory. This paper aims to map a QL model to what we understand as consciousness. Through utilizing spectral graph theory developed by Scholes, the existence of classical systems that produce QL states is proven via k -regular graphs that model networks. Aspects of these graphs are then translated to Khrennikov's QL formalism for neural behavior. Next, measurable neurophysiological quantities associated with consciousness are discussed within the context of relating these measurables to the theories and results posed by Khrennikov and Scholes. Finally, experiments linking the QL models with neurophysiology are proposed.

This paper represents my work in accordance with University regulations.

Natasha Greenstein

1 Introduction

While scientific discovery has progressed leaps and bounds in the past century, two concepts seem to continuously frustrate yet inspire researchers, students, and curious humans alike: consciousness and quantum theory. Born in the previous century, quantum theory is merely a child relative to the study of consciousness, which has interested countless for millennia. From the ancient Chinese Yijing, or The Book of Changes which divides consciousness into three layers: cosmological consciousness, consciousness of the human self, and political consciousness [1] to Descartes's idea that consciousness and matter could either live on the same or separate realms, the question of existence has inspired humanity for as long as written history allows for. [2].

Future science will comfort us in the notion that human existence is an incredibly improbable phenomenon, and begs the question of how, within our universe that (in general) requires entropy to increase, energy consolidated in such a way which birthed consciousness? As modern medicine progressed, it has become generally accepted that the brain plays a major role in the manifestation of consciousness, thus providing a physiological subject to study. Yet, even with the most modern imaging techniques, such as functional magnetic resonance imaging (fMRI) and electroencephalograms (EEG), we are still nearly as perplexed by the human experience as infamous philosophers.

Centuries after Descartes' grapple with consciousness, a new topic of perplexity emerged. On December 14, 1900, ignited by Max Planck's atomic energy quantization formula, quantum theory was born. Planck bore no explanation for this formula's discrete nature, yet it fit his empirical data beautifully [3]. As infamously declared by quantum legend Richard Feynman, "if you think you understand quantum mechanics, you don't understand quantum mechanics" [4]. Students' first encounter with quantum mechanics is notoriously convoluted and confusing – expectedly so. Quantum physics disrupts one's entire notion of the innate behavior of the universe. It makes one question the very fabric of reality and is unbearably counterintuitive.

Yet, instead of fearing quantum theory's weirdness, a few unconventional quantum enthusiasts dared to wonder whether one strange, spooky concept may be used to explain another. Ideas that some sort of quantum theory could potentially describe consciousness propagated through discussions heavily in the 60s and 70s, often with peculiar origins. In *How the Hippies Saved Physics*, David Kaiser tells the story of UC Berkeley's "Fundamental Fysiks Group" - an influential theoretical physics discussion group born out of department budget cuts and metaphysical curiosity. Lost during post-WW2's era of prioritizing brute force calculations and weapons manufacturing, the philosophical implications of quantum theory reemerged within these discussion groups; "many of the ideas that now occupy the core of quantum information science once found their home amid an anything-goes counterculture frenzy, a mishmash of spoon-bending physics, Eastern mysticism, LSD trips, CIA spooks chasing mind-reading dreams, and comparable 'Age of Aquarius' enthusiasms" [5]. Possibly not so surprisingly, physicists in altered states of con-

consciousness were some of the first contributors to the idea that quantum theory and consciousness may be intertwined. Curiously enough, psychedelics (known producers of altered states of consciousness (ASC)) alter brain wave oscillation frequencies [6] – which are regularly associated with ASC [7]. This is explored further in section 3.

The Bay Area psychedelic quantum enthusiasts were not the only ones to develop quantum theories of consciousness. The most notable and widely spread quantum-mind theory is Roger Penrose and Stuart Hameroff’s 1994 Orchestrated Objective Reduction Theory (Orch OR). Orch OR suggests that consciousness results from quantum computations occurring within the neuronal microtubules, leading to a form of quantum coherence and wavefunction collapse. This process is hypothesized to generate moments of conscious awareness, linking neural activity with fundamental principles of quantum mechanics [8]. However, Orch OR has valid critiques and limitations. Most glaringly, some of the physics is attributed to unknown effects of quantum gravity, for which a complete and universally accepted theory has not yet been established. There is also a lack of a clear explanation for how a hot, wet, noisy environment could retain true quantum-scale coherence on a necessary time scale.

An important function of Orch OR, however, is that it dares to merge the subjects of consciousness and quantum theory so that one may take a more interdisciplinary approach to fundamental questions. To explore the link between quantum theory and consciousness, we offer a different perspective; the idea that consciousness can be described by a quantum-like (QL) system – one that relies on the mathematics and nature (think entanglement and superposition) of quantum physics, yet crucially, operates on a classical level. To better comprehend QL systems, recall the analogies between mechanical, electrical, and rotational systems – all governed by the same equations of motion [9]. See Table 1 for reference. While the mathematics of these systems are the same, the physical realities are entirely different.

$$F = m \frac{d^2 X}{dt^2} + b \frac{dX}{dt} + KX \quad (1)$$

$$T = J \frac{d^2 \theta}{dt^2} + b \frac{d\theta}{dt} + K\theta \quad (2)$$

$$V = L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \left(\frac{1}{C} \right) q \quad (3)$$

In this paper, we propose that consciousness is to quantum theory as translational dynamics are to electrical or rotational dynamics – same mathematics, different plane of existence.

Table 1: Tabular representation of common classical analogous systems

Mechanical Translational System	Electrical System	Mechanical Rotational System
Force (F)	Voltage (V)	Torque (T)
Mass (M)	Inductance (L)	Moment of Inertia (J)
Damping Constant (B)	Resistance (R)	Torsional Damping Constant (B)
Spring Constant (K)	Reciprocal of Capacitance (1/C)	Torsional Spring Constant (K)
Displacement (X)	Charge (q)	Angular Displacement (θ)
Velocity (v)	Current (i)	Angular Velocity (ω)

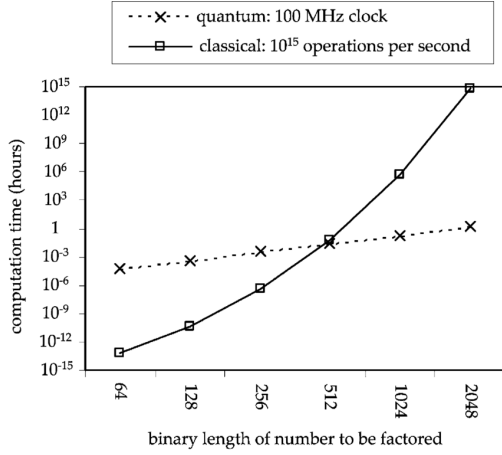
1.1 Motivations

The natural next step is to ask whether any classical systems exist that could potentially exhibit QL behavior under certain conditions. Scholes 2024 details systems that exhibit QL behavior, which I will closely focus on in this paper. These systems are consistent with [10] which I will also closely follow. Two open questions posed by Scholes: (i) is there a QL advantage for function? (ii) could QL states function and be used by neural networks in the brain? I hypothesize that the answers to questions (i) and (ii) are inextricably intertwined. In Michio Kaku’s *The Future of the Mind*, the idea of being able to increase intelligence by simply tinkering with the physicality of the brain is discussed [11]. Yet, for every quantity we could theoretically alter, there exists a tradeoff. For instance, increase of brain volume via neuron length extension, increase of signal speed via neuron thickening, and increase of neural connections all lead to greater energy consumption, rendering the body too hot to function, or signals too slow to communicate properly. It would follow that the capacity for intelligence of brains is physically limited by their ability to process information on necessary timescales, or in simpler terms: energy efficiency constraints limit physical alterations aimed at increasing intelligence (often associated with consciousness). If energy efficiency is the primary constraint unsatisfied by current classical approaches, in which framework could consciousness advantageously exist?

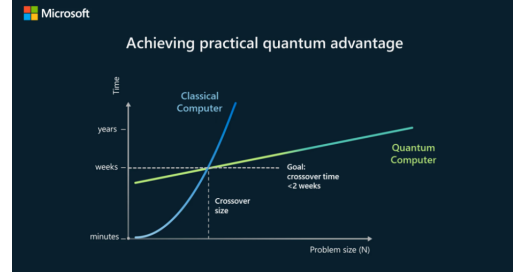
This is where quantum-like (QL) comes in. The incentive of QL formalism may be explained using the same logic that proponents of quantum computers use when discussing the quantum advantage in computing.

In these figures, we see that there is a particular crossover point where quantum algorithms are more efficient than classical algorithms. It has been demonstrated that while classical algorithms outperform quantum algorithms at simple tasks, complex optimization problems are more suitable to quantum formalism. We argue that due to the complexity of neural networks, consciousness “lives” on the right of the crossover point. I emphasize that “lives” is in quotations because while consciousness may be more efficiently *described* by *quantum* formalism, it is still understood to be a *classical* process.

An example that highlights similar approaches from a more technical side explores quantum-inspired tensor networks for more efficient machine learning, which is notoriously modeled after



(a) Caption for image 1



(b) Caption for image 2

Figure 1: Quantum vs. classical comparison of computation time in hours vs. numbers to be factored during computation. Left from MIT [12], right from Microsoft [13]

human cognition. Stoudemire and Schwab demonstrate that “mapping [classical network] data into an extremely high-dimensional Hilbert space is likely advantageous for producing models sensitive to high-order correlations among features,” such as the brain. They conclude that “there is great promise in investigating the power of quantum-inspired tensor networks for many other machine learning tasks” [14]. Should consciousness indeed follow a QL formalism, neural signaling would be subject to benefits of quantum information theory that appear reminiscent of satisfying energy efficiency constraints, such as those in [14]. One such notorious advantage is called superdense coding. The simplest illustration of super-dense coding is the transfer of two classical bits via one entangled qubit.

1.2 Superdense coding

Imagine a scenario where Alice and Bob initially share an entangled qubit. Along with Alice’s initial knowledge of her part of the qubit, she is also in possession of two classical bits of information that she intends to send to Bob. These classical bits modulate the quantum operations to be done on the shared qubit, i.e. instructions. Bob may only access the qubit and the after effect of operations acting on their shared qubit. Because Alice has two classical bits, she applies two arbitrary gates (Pauli gates: I, X, Y, Z) to the one entangled qubit. Bob then knows which gates are applied by examining his part of the qubit. By combining this knowledge with his known information about the qubit’s initial state, Bob is then able to decode which gates were applied, and since the instructions for which quantum gates were to be applied corresponds to classical information, Bob is then aware of the two classical bits of information through the exchange of only one quantum bit of information.

Extrapolating superdense coding to n entangled qubits, one may transmit $2n$ classical bits of information [15]. This is one of the key properties quantum computers aim to exploit for efficiency purposes. QL formalism can also utilize properties such as superdense coding, providing a potential link to the energy efficiency constraints on physical changes to the brain's structure.

2 Mathematical formulation

2.1 Graph theory

In classical neuroscience, a neuron is the base unit by which information is processed. The average adult human brain is composed of about 100 million neurons [16], forming various complex networks. These networks, defined as neural networks (not to be confused with neural networks of machine learning), have contributed immensely to modern information theory [17], and have been studied in efforts to understand consciousness [18]. In most of these discussions, network dynamics are based in classical probability theory. Unfortunately, classical probability theory has failed to concretely describe consciousness as it has been observed today. Additionally, proposed theories of quantum consciousness exist in the literature [19,20].

However, as discussed above, these actual quantum theories fail to explain certain issues such as decoherence in a hot, wet environment, and relied on unexplained “quantum gravity” effects. Yet, classical dynamics still come up short. As such, *employing* quantum theory may still offer an intuitive solution to the remarkably similarly mysterious problem of consciousness. Thus, the idea of QL is born, and we attempt to map the formalism of quantum theory to network dynamics on a classical scale. The first question is, are there classical systems that exhibit QL behavior, and can networks be represented by these systems? The answer to both is yes, and these systems present themselves in particular types of network models: graphs.

The graphs in question are special because they satisfy the axioms that map a classical process to quantum theory:

1. Individual graphs produce a distinct eigenstate from the rest (emergent state)
 - a. Quantum analog: basis vectors of wavefunction
2. Coupled subgraphs produce a distinct superposition eigenstate (emergent state)
 - a. Quantum analog: literally superposition
3. Graphs are robust to noise as well as to frequency fluctuations of the oscillators (nodes)
 - a. Not an analog, more so this system may exist on a classical scale
4. The QL bit needs to comprise a basis of two distinguishable ‘on’ states to produce superpositions by interacting with other networks of oscillating dipoles.

- a. Quantum analog: spin up spin down in electron
- 5. Unitary operations on graphs and subgraphs should be defined
 - a. Quantum analog: unitary operations preserve trace/probabilities, base of quantum information theory and quantum computation

As outlined in Scholes 2024, examples of graphs that exhibit this behavior are k -regular (every node connects to k edges), random (probability p of edge connection) graphs. Imagine a network of neurons N described by coupled k -regular, random graph G with n nodes and m edges. Containing two layers of dipole oscillation dynamics, the first layer of dipole oscillation behavior is exhibited by the nodes of this graph. Each of the n oscillating nodes may flip between two orthogonal states. The m edges are signed as before (positive or negative), indicating direction flow of information. Now that the network is described as a graph, we may analyze it. Graph G can be represented by an adjacency matrix A . An adjacency matrix describes connections between the different nodes in the graph. For these types of graphs, the eigenvalues of adjacency matrix A , also known as the spectrum of G , characterize the emergent states of graph G . Emergent states (eigenstates) are the eigenvectors of adjacency matrices that are distinctly separated from the bulk of the spectrum, thus rendering these states robust to noise – an issue consistently raised with actual quantum theories of consciousness. Figure 2 illustrates this phenomenon.

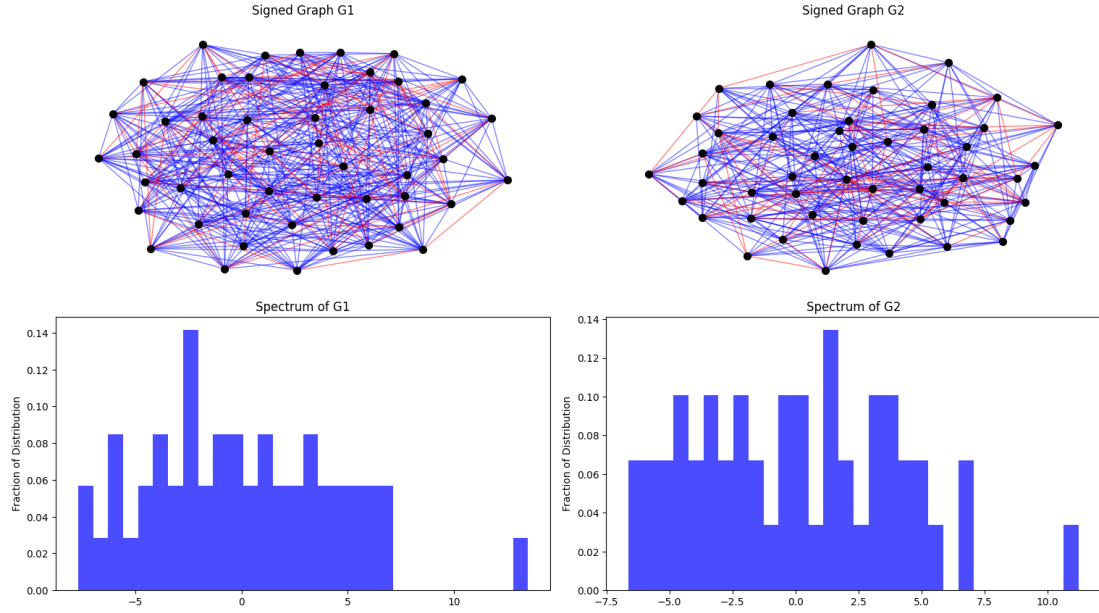


Figure 2: Top: k -regular, random graphs $G1$ and $G2$ with $n = 50$ nodes, $k = 20$ connections per node, $P(\text{positive (red) edge}) = 0.2$, $P(\text{negative (blue) edge}) = 0.8$, number of edges removed $G1 = 20$, $G2 = 100$. Bottom: spectrums of $G1$ and $G2$. Isolated bar = eigenvalues

In Figure 3, we explore the relationship between spectra eigenvalues and edge signage. The bottom graphs nicely demonstrate that flipping the sign of edge connections flips the sign of the eigenvalues.

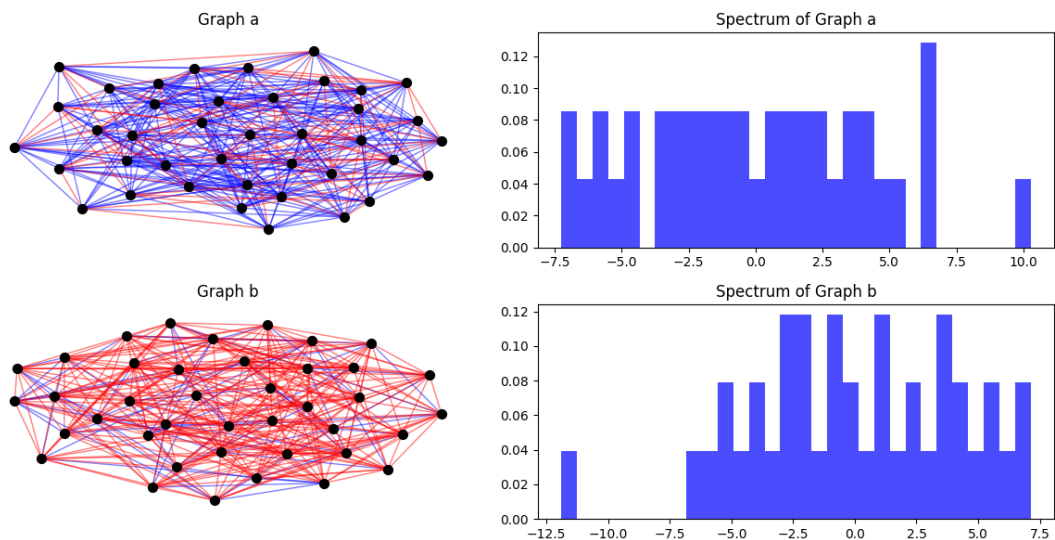


Figure 3: Top: k-regular, random graphs G_a and G_b with $n = 50$ nodes, $k = 20$ connections per node, $G_a: P(+ \text{ (red) edge}) = 0.2$, $P(- \text{ (blue) edge}) = 0.8$, $G_b: P(+ \text{ (red) edge}) = 0.8$, $P(- \text{ (blue) edge}) = 0.2$, number of edges removed $G_a = 20$, $G_b = 100$. Bottom: spectrums of G_a and G_b . Isolated bar = eigenvalues

Next, imagine graph G is comprised of two subgraphs G_{a1} and G_{a2} , each coupled k-regular random. Edges are signed as before. The adjacency matrix A is simply the tensor product of G_{a1} 's adjacency matrix with G_{a2} 's, and its eigenstates are just linear combinations of the eigenstates of the subgraphs, G_{a1} and G_{a2} . The second layer of dipole oscillation behavior is exhibited when two coupled subgraphs interact, as G can flip between its two eigenstate linear combinations of subgraph eigenstates. Figure 4 illustrates this QL superposition phenomenon nicely.

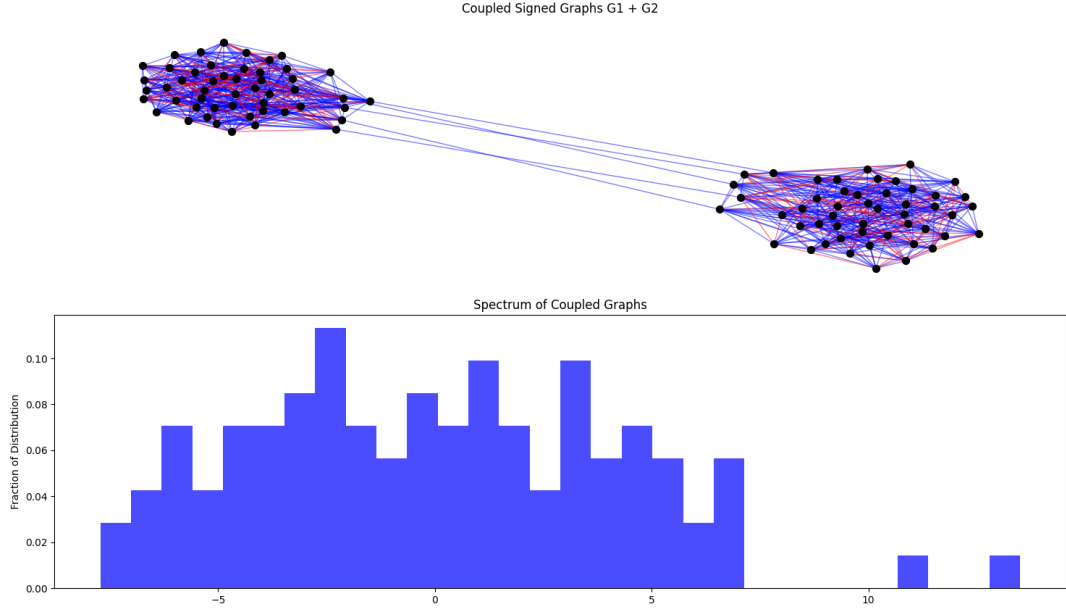


Figure 4: Top: coupled k -regular, random graphs $G1$ and $G2$ with $n = 50$ nodes, $k = 20$ connections per node, $P(\text{positive (red) edge}) = 0.8$, $P(\text{negative (blue) edge}) = 0.8$, number of edges removed $G1 = 20$, $G2 = 100$. Number of edges added = 5, coupling is schematic. Bottom: spectrums of $G1$ and $G2$. Isolated bars = distinct coupled eigenvalues

The axioms discussed before are satisfied by these graphs as below:

1. The emergent states of G are linear combinations of the eigenstates of $Ga1$ and $Ga2$, thus producing superposition states.
2. The k -regularity and randomness of the graphs ensure that the basis states and their superpositions are very robust for networks of arbitrary complexity and may be identified distinctly by their emergent eigenvalue. This allows the QL process to exist on a classical scale
3. Both superposition states are “on.”

Physically, how may these graphs be interpreted? Can we connect their structures back to network neuroscience? The idea is presented below:

Table 2: Graph to system translation guide

Graph element	Dipole physics	Neural system
node	oscillating dipole (small)	neuron
edge	coupling	synaptic connection
graph	oscillating dipole (big)	neural network
emergent state	<i>crucially NOT the oscillator frequency</i>	‘decision’ state

Table 3: Quantum to QL translation guide

Quantum	QL
wavefunction	emergent state (ES)
entanglement	coupling (between graphs)
superposition	superposition of ES

In the quantum world, we say that particles are represented by wavefunctions, or a linear combination of states to form a state vector. In the QL world, we say that networks are represented by ES, or a linear combination of subnetwork ES to form a state vector. Same mathematical object, different scales of existence, i.e., an analogous system.

2.2 QL formalism

We have shown the existence of networks that exhibit QL behavior. To progress, we now dive into the QL formulation discussed primarily by Andrei Khrennikov and colleagues. Let's break this down into the two layers of oscillating dipoles: single neurons dynamics, and graph/network dynamics. I will take on a micro to macro approach, describing first the QL representation of neuronal information states, and second the QL representation of network information states.

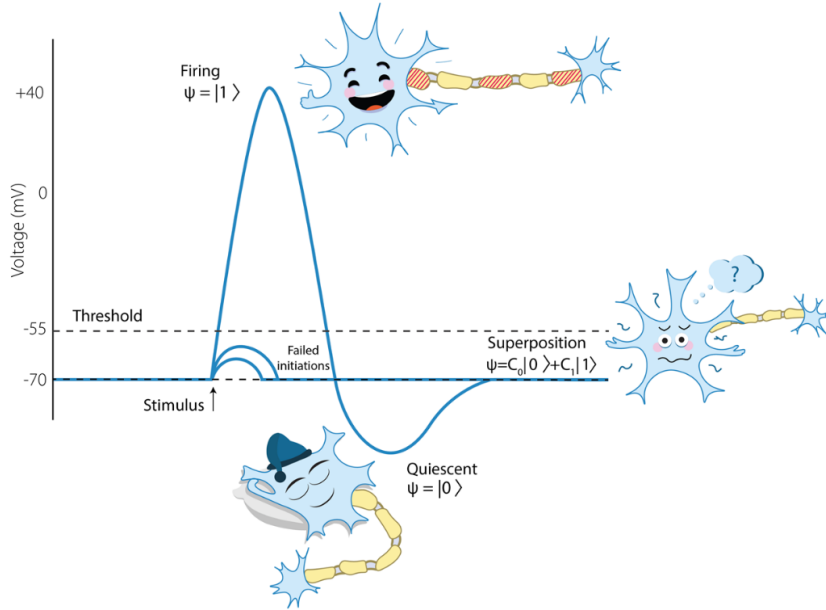


Figure 5: Physiological visualization of QL neuronal states. When translating to graphs, consider each neuron as an oscillating dipole that can flip between quiescent and firing states, and may also exist in the superposition state [10]

In quantum mechanics, the primary mathematical objects are wavefunctions, represented as state vectors, and operators, represented by matrices. When we get into quantum information theory, environments are introduced. Interactions with the environment are also represented by matrices. Technically, these could all be tensors, but for simplicity it is more digestible to think in terms of vectors and matrices.

In Figure 5, single neurons are represented with state vectors, which we remember from the previous section may also be thought of as oscillating nodes within a graph. Khrennikov asserts that “the superposition states represent uncertainty generated by the membrane potential V ” because “uncertainty in generation of action potentials is a consequence of stochastic operation of ion channels [10],” i.e. the main source of uncertainty is interactions with surrounding environment [21]. As such, potential membrane voltage measurements could tell us which state the neuron is in: Quiescent < -70 mV, Firing > -55 mV, and Superposition State $\in [-70$ mV, -55 mV]. Therefore, a QL single neuron depends on its openness to as an electrochemical system. To reiterate, these states would tell us the states of neurons within a network, not the emergent state (ES) of a network, which will be discussed subsequently. Neurons and networks both individually behave as QL systems, and the QL nature of single neurons contributes to the intrinsic behavior of networks as a whole.

If we assume thoughts or decisions are represented by emergent states, then *decision states* of networks could be *physically measured* by *emergent state values*. While Table 1 highlights the translations between graph theory and physical reality, Table 2 highlights the translations between the actual quantum world and QL, providing a basis for interpretation.

Table 4: Breakdown of Khrennikov formalism consistent with Scholes graphs

Variable	Physical Interpretation
G	Group of neurons, network
F	Psychological function to be performed by network
E	Surrounding electrochemical environment
$\rho(t)$	Decision state (emergent state) of G performing F in E

Table 4 is discussed next. The process of decision making may be considered decoherence of the QL network state of G in the basis of eigenstates of F . A decision state ρ is a steady state of the dynamics $\rho(t)$ of network represented by G . ρ is diagonal in the F -basis and represents the weights of possible F outputs. This makes sense considering the network ‘collapses’ into a state in the basis comprised of eigenstates of F .

ρ exists as a steady state due to interactions with the environment E . Even though neuron network G is technically trying to complete function F , interacting with the environment E is critical to convergence of G to steady state ρ .

A matrix is diagonal in a basis of its eigenvectors. Therefore, ρ , the steady state of G ,

has the same eigenvectors as F . Thus, we may say that when the network G reaches steady state ρ with the same eigenvectors as function F due to interactions with the environment E , the network has *completed function F* .

2.3 Thought experiment simulation

To illustrate this concept, let's conduct a thought experiment. Imagine you have an experience with dogs that causes you fear when you see one. This experience is encoded into environment E in some neurophysiological capacity, perhaps a more active amygdala. Let's call network G one that deals with associating objects with emotions. Function F is the network deciding whether to send a fear signal when network G deals with an object. Now, let's imagine you observe a dog. Because of past experience, your brain associates the dog with a sense of fear. This is reflected in the steady state ρ that G converges to when performing signal transmission function F under the context of past trauma encoded into environment E , which probably assigned higher weights to the state of "send signal" in this scenario.

Relating this formalism back to Scholes' graphical interpretation, network G , function F , and environment E all contribute to the network dynamics, which results in emergent state ρ to be measured. We can simulate this using the same base program that produced the earlier figures with clearly and specifically defined parameters.

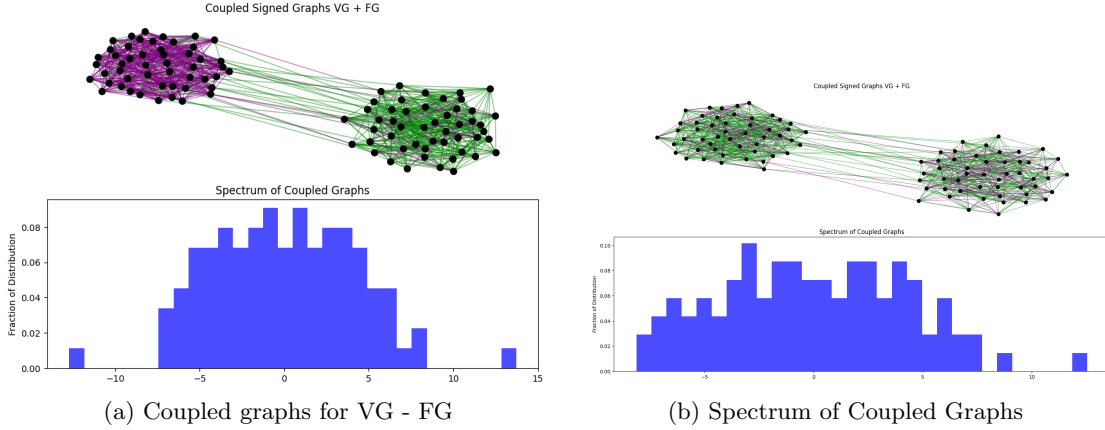


Figure 6: Coupled graphs and their spectra for fear experiment. $G1$ = visual graph (VG), dealing with level of perception. $G2$ = fear graph (FG), dealing with level of fear. 'Positive' = green = perceiving/fear. 'Negative' = purple = ignoring/no fear. Edges removed from VG = 20, edges removed from FG = 100, removed connections = 25. LEFT: negative edge fraction = 0.2 (VG), 0.3 (FG), 0.1 (coupling). RIGHT: negative edge fraction = 0.8 (VG), 0.1 (FG), 0.1 (coupling).

In Figure 6, we deal with two scenarios. In the first (left), we have a situation where this subject perceives the dog heavily and is also quite scared of it. The spectrum yields two distinctly maximal eigenvalues, which would be associated with a decision to react to the dog. Maybe jump a little or let out a yell. In the second (right), we have a situation where the subject does not

perceive the dog very much, yet holds great fear regarding dogs. This time, the spectrum yields a distinctly maximal and minimal value. These results would map to two distinctly different actions: visceral reaction if the fear graph ‘takes over’ and the subject happens to notice the dog, or non-reaction if the visual graph ‘takes over’ and the subject does not notice the dog.

From Figures 2-4, we can clearly observe the emergent state eigenvalue’s distinctness from the rest of the eigenvalue distribution. This distance is what signifies these emergent states as important, so it is worth it to explore how it changes with graph parameters. These simulations are discussed next.

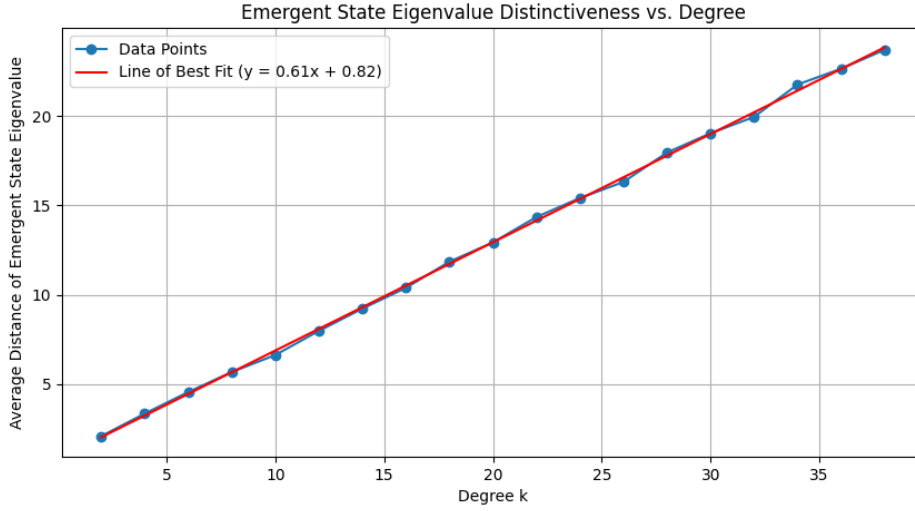


Figure 7: Average Distance of Emergent State Eigenvalues vs. k -Degree. 2 coupled k -regular random graphs G1 and G2. $n = 100$ nodes, trials per $k = 10$, $k = [2, 40]$, removed edges = 20 (G1), 100 (G2). Number of connections = 5, negative edge fraction = 0.2. In taking averages, emergent states were excluded. Line of best fit calculated using python’s ‘polyfit’ function.

Figure 7 demonstrates that as k increases, the distinctness of the emergent states increases linearly. Translating this result to reality, we could say that as the complexity of a neural network increases, so does the separation of a determined ‘thought’ compared to noise. This makes sense if you extrapolate the result to what we know from current neuroscience and frankly just life; the more a brain learns, processes, and builds synaptic connections, the easier it becomes to make confident, distinct decisions.

An important caveat to note regarding Figure 7, is that the graphs are not *technically* k -regular. Because we have introduced negative edges, the k -degree is ‘decreased’, as k is only determined by the number of positive edges per node. To clarify, there are still k edges connected to each node, just some are positive, and some are negative. For typical k -regular graphs, we expect the slope to be equal to 1 [22]. However, the deregulation due to signage presents irregular

results. This concept is further explored in Figure 8, for which all graph parameters are the same, except the negative edge fraction, which has increased from 0.2 to 0.9.

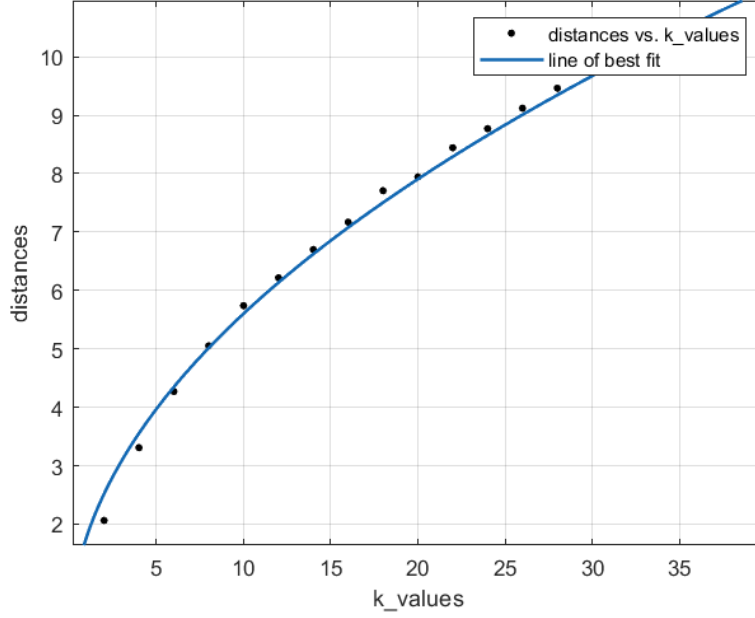


Figure 8: Average Distance of Emergent State Eigenvalues vs. k-Degree. 2 coupled k-regular random graphs G1 and G2. $n = 50$ nodes, trials per $k = 10$, $k = [2, 40]$, removed edges = 20 (G1), 100 (G2). Number of connections = 5, negative edge fraction = 0.9. In taking averages, emergent states were excluded. Model fitted using MATLAB's CurveFit 'power' function.

Here, the data best fits $y = 1.78x^{0.48}$, or approximately a square root relation. Supposing negative in this instance represents the opposite direction of neuron information transmission, the results of Figure 8 suggest that as neural circuits reverse their wiring, the ability to produce a distinct state diminishes. This result is less easily translatable to a physiological phenomenon, and requires more careful examination.

Refocusing on the linear relationship in Figure 7, we may compare Figure 7 results to the Alon-Boppana result, which states that the distance between the maximal/minimal and next eigenvalue of k-regular graphs is:

$$\lambda_1 - \lambda_0 = k - \sqrt{2(k-1)} \quad (4)$$

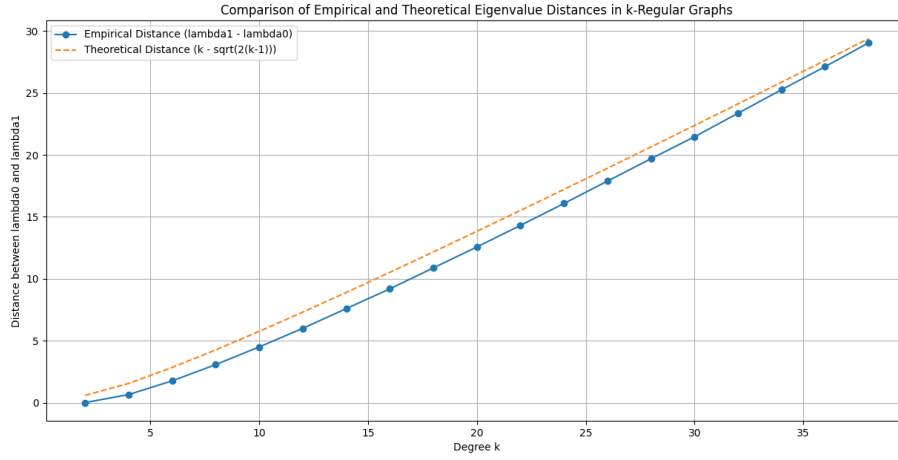


Figure 9: Average Distance Between First Two Eigenvalues vs. k . $n = 100$ nodes, trials per $k = 100$, $k = [2, 40, 2]$. Blue = empirical data, orange = theoretical

Figure 9 verifies that the graphs in question behave as anticipated. Intuitively, this validates the results from Figure 7 by confirming an essentially linear relationship between eigenvalue distances.

Next, the coupling between graphs compared to eigvalue distinctness is explored.

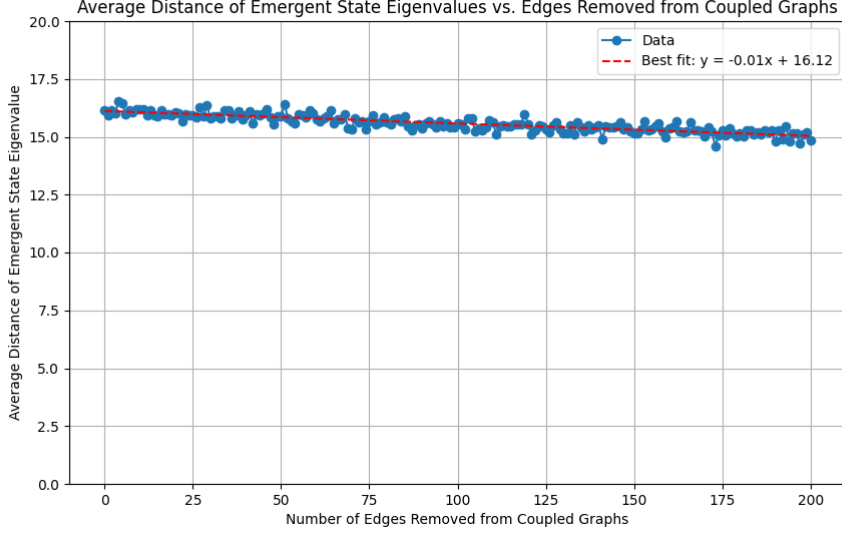


Figure 10: Average Distance of Emergent State Eigenvalues vs. Number of Edges Removed. 2 coupled k -regular random graphs $G1$ and $G2$. $n = 40$ nodes, $k = 20$, removed edges = 20 ($G1$), 100 ($G2$). Number of connections = 200, negative edge fraction = 0.2. In taking averages, emergent states were excluded. Line of best fit calculated using python's 'polyfit' function.

Figure 10 demonstrates that as the number of edges between graphs increases the distinctness of the emergent states decreases slightly. This may be explained best by relating these results first to quantum theory, and then to reality.

In quantum mechanics, relatively small interactions may be modeled using perturbation theory. Due to the linearity of quantum mechanics, one may add terms to the Hamiltonian that behave linearly, which allows us to individually examine the effect of these terms on the energy eigenvalues.

Perturbation theory is the analysis after the addition of small effect terms to the Hamiltonian. This theory may be represented by considering a system with the Hamiltonian given by

$$\hat{H} = \hat{H}_0 + \varepsilon \hat{C}, \quad (5)$$

where $\hat{H}_0$ is the Hamiltonian of the unperturbed system (decoupled graphs), ε is a small parameter (coupling degree), and $\hat{C}$ describes the perturbation that represents *intergraph* coupling interactions. We assume that the perturbation C is time independent.

Then, plugging this Hamiltonian into the Schrödinger equation:

$$\hat{H}\rho = E\rho. \quad (6)$$

Since ε is small, we can use a Taylor series expansion ε :

$$\rho = \rho^{(0)} + \varepsilon \rho^{(1)} + \varepsilon^2 \rho^{(2)} + \dots, \quad (7)$$

and by linearity,

$$E = E_0 + \varepsilon E_1 + \varepsilon^2 E_2 + \dots \quad (8)$$

Note that here E denotes *energy*, not the electrochemical environment as discussed previously.

Remember, since ε is small, the changes in ρ are small. These changes would be reflected within the graph spectra is a tiny gap between two emergent states ρ . The key point of the results from Figure 9 indicate that the number of neuron subnetworks coupled does not have a significant impact on the distinctness of an emergent state. The main contributions to an emergent state distinctness are from *intra* rather than *inter*-graph configurations. A question of further exploration may be something like altering which graph parameters to what extent will change the balance of influence between intra and inter-graph configurations?

In this paper, the next exploration is distinctness versus number of graphs coupled.

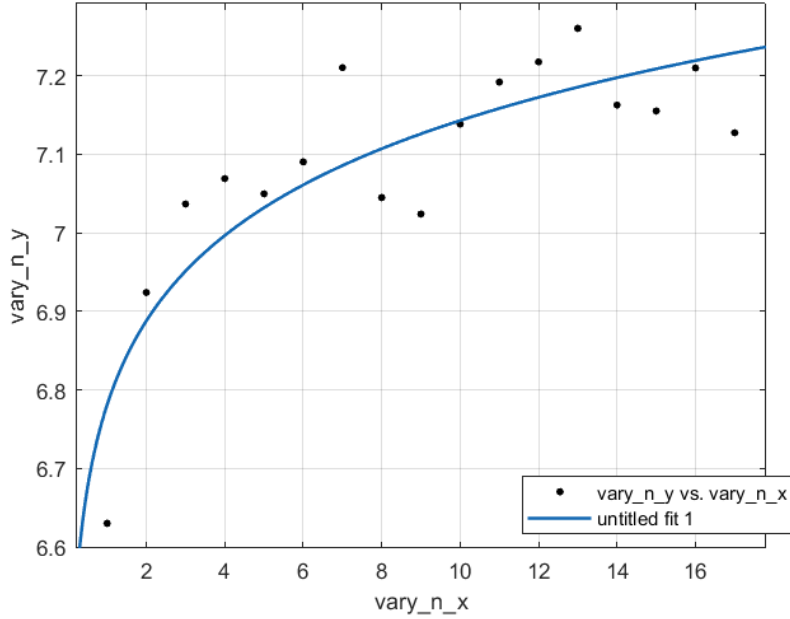


Figure 11: Average Distance of Emergent State Eigenvalues vs. Number of Graphs Coupled. N coupled k-regular random graphs, $N = [1, 18]$. $n = 30$ nodes, $k = 10$, removed edges = 20 (G1), 100 (G2). Number of connections = 5, negative edge fraction = 0.2. In taking averages, emergent states were excluded. Line of best fit calculated using python's 'polyfit' function.

Figure 11 demonstrates that as the number of coupled graphs increases most like $y = 6.67x^{0.02}$. The R-square value is 0.7390, as compared to a linear relationship R-square of 0.5015. This result again checks out with intuitive reality. We see that as more subnetworks join the total network, the distinctness of thoughts improves. Communication between different brain regions is generally associated with higher states of consciousness [23], thus providing a realistic basis for extracting more unique, confident thoughts. The next section of this paper will focus on methods to map these QL theories to neurophysiological observables, in an attempt to physically solidify a QL theory of consciousness.

3 Neurophysiological background

I will begin this section by restating the two questions posed by Scholes: (i) is there a QL advantage for function? (ii) could QL states function and be used by neural networks in the brain? I will answer these questions in tandem, as there is a hypothesis that the answer to question (i) provides insight into question (ii). To approach these questions, it seems natural to try to map neurophysiological processes to the previously defined formalism. First, a few neuroscience concepts must first be addressed. The main physical measurables central to the later proposed experiments are membrane potentials (previously discussed), brain wave oscillation frequencies, and cerebrospinal fluid.

3.1 Brain wave oscillations

Oscillations have been a theme of this paper. So, let's introduce one more kind: brain wave oscillations (BWO). BWO describe the rhythmic, fluctuating patterns of information transfer (electrical activity) between neurons. These oscillations are categorized into several bands of frequency ranges – each associated with ASC: delta waves (< 4 Hz) with sleep and unconscious processing, theta waves (4-8 Hz) with drowsiness, light sleep, or deep meditation, alpha waves (8-12 Hz) with relaxed, wakeful states, beta waves (12-30 Hz) with active, alert engagement with the external world, and gamma waves (30-80 Hz) with higher states of conscious perception [24–27].

BWO are thought to encode information primarily in two ways: temporal dynamics (timing of neural firings) and spatial coherence (synchronization of activity across brain regions). This dual encoding mechanism allows the brain to efficiently process complex information, from sensory perception to higher-order cognitive functions [23].

Frequencies of oscillations can influence information processing and integration. For example, theta and gamma oscillations have been observed to interact in a nested manner, where the phase of theta waves modulates the amplitude of gamma oscillations. This interaction is crucial for memory functions and provides insight as to how the timing of neural activity may encode specific information. This is consistent with the idea that oscillator frequencies play a role in the construction of our QL system [23].

Coherence across various brain regions suggests that oscillation activity is synchronized over large neural networks, which facilitates communication between distant brain areas. This synchronization is essential for integrating sensory inputs, coordinating motor outputs, and maintaining cognitive states. Studies have shown that during sensory processing and attentional tasks, gamma oscillations are correlated with coherent activity across the cortex [23].

In section 2, we discussed emergent states (ES) of networks/graphs G . These ES behave like wavefunctions in our QL theory, as such they have frequencies, amplitudes, and other properties of waves. Above, we discussed how brain wave oscillation frequencies are associated with ASC. The question is: are brain wave oscillation frequencies physically related to network dynamics - and subsequently, emergent states?

3.2 Cerebrospinal fluid and surrounding electrochemical environment

The second concept I will provide background on is cerebrospinal fluid (CSF), as a working hypothesis is that CSF modulates the membrane potential uncertainties of these QL states due to the composition of and interactions with interstitial ions, thus modulating consciousness and cognition. Firstly, what is CSF? CSF is a fluid that surrounds the brain and spinal cord. Current literature defines CSF's functions as providing mechanical support, protecting the brain and spinal cord from trauma, supplying nutrients to nervous system tissue, and removing waste products from cerebral metabolism. However, there are also studies that link changes in CSF to ASC, thus potentially linking CSF patterns to consciousness.

This idea is born out of research conducted at the Max Planck Institute of Self Dynamics and Organization in Göttingen, Germany in the summer of 2022 under Shobha Kapoor. We studied CSF flow as it relates to cilia and phenotypical expressions. Kapoor noticed that CSF flow patterns are consistent with changes in phenotype expression, proposing that cilia are not in fact responsible for information transport. This opens an avenue for CSF to be a means of information transport.

In [28], it was concluded that “patients with lower CSF exhibited significantly worse memory, visuo-constructive capacities, and verbal fluency. CSF alterations could contribute to some of the cognitive deficit observed in patients with Alzheimer’s Disease. Diagnosis and treatment of CSF alterations in geriatric patients with neurocognitive disorders could contribute to the prevention of their cognitive decline.” This study provides a direct link between CSF quantification of some measure and cognitive function.

Another study from 2013 examined the relationship between respiration and CSF determined that CSF flow changed with respiratory rates [29]. This again establishes a link between breathing/meditation and changes in CSF. Crucially, there already exists an established link between breathing/meditation and ASC. Thus, we see a link from CSF to ASC once again. This study is curiously reminiscent of results out of Trinity College Dublin that explore entangled

behavior between heart and brain protons [30].

These studies highlight our CSF motivation. Now, we must figure out where CSF fits in the QL formalism. The composition of CSF may hold a key. CSF contains the ions such as Na+, K+, Ca2+, and Cl- the ions that regulate action and membrane potentials:

1. Sodium (Na+) flow in and depolarizes neurons so that an action potential may be triggered
2. Potassium (K+) flows out of a neuron once the action potential peaks (neuron fires)
3. Calcium (Ca2+) flows in to depolarize neurons such that neurotransmitters are released into the synaptic cleft, i.e. interneural communication
4. Chloride (Cl-) ions determine the negativity and therefore excitability of neurons. since they are negatively charged, the more internal neuronal Cl-, the less excitable.

CSF directly influences the ionic balance and environment surrounding neurons. It is straightforward then that CSF also directly influences neuronal membrane potentials. As discussed above, Khrennikov’s 2018 QL model for a single neuron “is based on quantum-like representation of uncertainty in generation of action potentials. This uncertainty is a consequence of complexity of electrochemical processes in the brain; in particular, uncertainty of triggering an action potential by the membrane potential.” The word that sticks out is uncertainty – a fundamental principle in quantum mechanics.

In QM, uncertainty is defined by $\Delta x \Delta p \geq \frac{\hbar}{4\pi}$,

where x denotes position, p denotes momentum, and $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$

and $\langle f(x) \rangle = \int \psi^*(x) f(x) \psi(x) dx$ where $\psi(x)$ is the wavefunction in position space

and p follows the same formulation.

Uncertainty in QM is directly related to the wavefunction (and thus probabilities of certain outcomes). It would follow that uncertainty in QL is also defined by these quantities and formulas. should this be the case, CSF can be linked to a modulator of QL wavefunctions, as determined by changes in uncertainty. From Table 3, we see that changes in CSF would thus be associated with changes in electrochemical environment E . Crucially, CSF is a classical object, which is consistent with the idea of QL. If CSF patterns are correlated with cognition/consciousness and are known to directly influence aspects of our QL theory, then *through detecting changes in CSF patterns, which theoretically contribute to environment E , we could use these ASC measurements to construct our QL to neurophysiology map.*

4 Experimental proposals and further questions

If we can perform an experiment that demonstrates neural behavior consistent with the graphs modeled by Scholes and the formalism derived by Khrennikov, we will be able to provide evidence

that consciousness (in order to adhere to energy efficiency constraints) and potentially other systems besides particles, are governed by the laws of quantum theory. A potential experiment is outlined below, with motivations rooted in the fact that psychedelics are known to produce ASC, and known to alter BWO [31].

The primary objective of this experiment would be to map neurophysiological quantities to quantum-like (QL) description of consciousness/cognition I propose accomplishing this through induce altered states of consciousness via a variety of methods such as psychedelics, meditation/breathing exercises, anesthesia (unconscious state), and of course a sober state for a conscious control. The researchers will then assign human subjects a task (or multiple tasks) in each state: one problem solving task, one memory recollection task, and/or one emotional analysis task. Should any of these tasks be too complex, they may switch to a watching/listening task. In these trials, one would have to consider Question Order Effect (as proposed by Khrennikov) by repeating experiments in various orders.

During data collection, BWO, CSF flow, and action/membrane potential dynamics would be measured for the reasons outlined above. Data analysis would consist of following Khrennikov's formulation of quantum-like consciousness, attempting to introduce CSF flow to environmental variable E , attempting to fit data patterns to formulation, and discussion of data gaps.

A hypothesis is that CSF flow and psychological task contribute to environment E , potential dynamics and psychological task contribute to function F , and brain wave oscillation frequencies contribute to neuron groups G . Or, they all contribute to all E, F, G and ρ and we need to figure out how. In the discussion of this experiment, it would also be worthwhile to try to map a classical version of the dynamics as well, so that to address the energy efficiency advantage question, one may compare energy transmission calculations for classical and QL frameworks.

If consciousness truly behaves as an analogous system to quantum theory, this has massive implications for medicine, technology such as artificial intelligence, and our understanding of nature. One begins to ask questions that pierce the fabric of reality, such as whether telepathy is possible. As such, I propose a series of questions about QL formulation that could be answered with similar experiments:

1. Physical Basis for QL State
 - (a) What is meant by the superposition state?
 - (a) What are the critical points?
 - (b) When/why does the 'wavefunction' (using quotes b/c QL) collapse?
2. How can we detect function from interacting with QL states? What should we look for?
 - (a) Energy efficiency (EE) principle

- i. Does the wavefunction collapse when it is energetically efficient to do so?
 - i. What is the EE threshold?
- (a) How does the state vector transform under altered states of consciousness
- (a) How do QL states interact? Can it happen over long range?
- 3. Is telepathy possible?
 - (a) If one could entangle two minds before an ES is measured, and assuming an ES is a thought, could the two minds essentially behave like a classic Alice and Bob information sharing problem? i.e., could they come to different sides of the same thought?
 - (b) Assume telepathy as described above is indeed possible. If universal consciousness exists, is it technically possible to communicate telepathically with *anyone*? What about non-human conscious beings?
 - (c) Do psychedelics make it easier to do so?

5 Conclusion

As a physicist – especially one interested in the quantum world, one is often forced to confront the very fundamental mysteries of the universe. As a human, one is forced to confront quite a significant fundamental mystery every single day: the human experience. This paper aims to connect the two. Through the lens of QL systems, we’ve embarked on a journey that attempts to marry neurophysiological phenomena with a robust QL theory, suggesting a novel interpretation of consciousness.

Using Scholes’ graphical models that are consistent with Khrennikov’s QL formalism and actual measurable values such as membrane potentials, BWO and CSF flow, a QL framework has been derived, and experiments have been proposed to fit experimental data with theory.

As we conclude, the entanglement of QL theory and consciousness research pushes the frontier, seeping with motivations for further inquiry and experimentation. The proposed QL model of consciousness, grounded in the fabric of neural network dynamics and altered states of consciousness, opens reveals unique pathways for understanding the mind’s labyrinth. It challenges us to reconceptualize not just consciousness, but systems as a whole. What other systems exhibit QL behavior? Results from these experiments would promise not just a deeper understanding of the mind but also revolutionary implications for neuroscience, artificial intelligence, and elements that may bring the realm of science fiction ever closer to reality.

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